

SANDRO PAPAIS

Ph.D. Candidate | Machine Learning for 3D Perception & Robotic Autonomy

✉ sandro.papais@robotics.utoronto.ca ☎ 514-979-2593 📍 Toronto, ON, Canada
🌐 github.com/sandropapais 🔗 sandropapais.github.io in linkedin.com/in/sandropapais

ROBOTICS & ML EXPERIENCE

Robotics & Machine Learning PhD Researcher

University of Toronto | Robotics and Artificial Intelligence Laboratory

📅 09/2021 – Present

📍 Toronto, Canada

- Advancing spatiotemporal representations for efficient robotic autonomy across perception, prediction, and planning.
- Developed CVPR/ICCV/ICRA-published multimodal transformer architectures for 3D detection, tracking, forecasting, and end-to-end driving.

🗣️ “U. of T. researchers are making self-driving cars safer.” U. of Toronto (2024).

Machine Learning Perception Researcher

Zoox

📅 05/2025 – Present

📍 San Francisco, California

- Developed sparse-attention transformer for 3D perception. Patent filed.
- Developing joint multimodal perception-prediction transformer architecture with temporal reasoning for real-time autonomous driving.
- Supporting data curation, validation, latency optimization, and deployment.

Autonomy Software Engineer

NGC Aerospace

📅 11/2019 – 08/2021

📍 Sherbrooke, Canada

- Developed and validated visual navigation software for Firefly Aerospace's Blue Ghost lunar lander and Laflamme Aero's LX300 eVTOL.

🗣️ “Made in Canada, tested on the Moon.” Government of Canada (2025).

Visiting Robotics Researcher

NASA Jet Propulsion Laboratory

📅 07/2019 – 10/2019

📍 Pasadena, California

- Developed 3D visual navigation algorithms and mission design for an autonomous smallsat asteroid-rendezvous mission (2020 IEEE Aerospace).

Robotics Researcher

European Space Agency & Reaction Dynamics Lab & McGill Univ. DECAR Lab

📅 05/2018 – 06/2019

📍 Netherlands & Canada

- Developed visual localization & 3D mapping algorithms for planetary robotics.
- Researched launch-vehicle GNC and authored \$500k R&D funding proposal.

Robotics & ML Teaching

Perception for Robotics Teaching Assistant

University of Toronto

📅 09/2025 – 12/2025

📍 Toronto, Canada

- Designed and graded course assignments and research projects on deep learning, 3D vision, vision-language models, and neural scene reconstruction.

Self Driving Cars Course Developer

Coursera

📅 05/2022 – 08/2022

📍 Toronto, Canada

- Developed ML-based perception, localization, and control modules using modern deep learning methods for 120k-learner autonomous driving MOOC.

PROFILE

Robotics and machine learning researcher with 8 years of R&D experience in autonomous systems. Research focuses on spatiotemporal representation learning, vision-based 3D perception, sensor fusion, forecasting, planning, efficient transformers, and real-time robotic autonomy.

EDUCATION

Ph.D. Robotics & Aerospace Engineering

University of Toronto

📅 August 2026

- Advisor: Steven L. Waslander
- Thesis: Spatiotemporal reasoning and representations for robotic autonomy.
- Cumulative GPA: 3.97/4.00.

B.Eng Honours Mechanical (Thesis)

McGill University

📅 May 2019

- Advisor: James R. Forbes
- Thesis: Interplanetary spacecraft navigation.
- Major GPA: 3.65/4.
- Deans Honour List 2014-15.

TOP AWARDS

🎓 Qualcomm Innovation Fellowship 2025
\$70k for autonomous driving foundation models research, out of 266 proposals.

🎓 Toronto Robotics & AI Fellowship 2021-26
\$100k for PhD research over 5 years.

🎓 Aerospace Institute Fellowship 2021-26
\$50k for PhD research over 5 years.

🎓 Molson Kenneth Fellowship 2025-26
\$9k for PhD research over 2 years.

🎓 Ontario Graduate Scholarship 2024
\$15k for spatiotemporal ML research.

🏆 SAE AutoDrive Challenge II 2022
\$28k for 1st/10 autonomous driving vehicles at international competition.

🏆 Spaceport America Genesis Cup 2018
1st/124 universities at international rocket engineering competition.

RESEARCH

Refereed Publications

- S. Papais, M. Antione, and S. Waslander (2026). "ReDrive: Rethinking Sparse Scene Representations for End-to-End Driving". *Under Review*.
- S. Papais, L. Feng, M. Jain, B. Rezaei, and S. Waslander (2026). "BeyondSight: Object Permanence for End-to-End Autonomous Driving". *Under Review*.
- L. Wang, M. Lavoie, S. Papais, ..., M. Pavone, and S. Waslander (2026). "Trends in Motion Prediction Toward Deployable and Generalizable Autonomy." *Foundations and Trends in Robotics*.
- S. Papais, L. Feng, C. Cossette, and L. Ge (2026). "SToRe3D: Sparse Token Relevance in ViTs for Efficient Multi-View 3D Object Detection". *Conference on Computer Vision and Pattern Recognition*.
- B. Cheong, ..., S. Papais, and S. Waslander (2026). "SCATR: Mitigating New Instance Suppression in LiDAR-based Tracking-by-Attention via Second Chance Assignment and Track Query Dropout". *Winter Conference on Applications of Computer Vision*.
- S. Papais, L. Wang, B. Cheong, R. Ren, and S. Waslander (2025). "ForeSight: Multi-View Streaming Joint Object Detection and Trajectory Forecasting." *International Conference on Computer Vision*.
- S. Papais, R. Ren, and S. Waslander (2024). "SWTrack: Multiple Hypothesis Sliding Window 3D Multi-Object Tracking." *International Conference on Robotics and Automation (ICRA)*.
- S. Wu, N. Amenta, J. Zhou, S. Papais, and J. Kelly (2023). "aUToLights: A Robust Multi-Camera Traffic Light Detection and Tracking System." *20th IEEE Conference on Robots and Vision (CRV)*.
- S. Papais, B. Hockman, ..., S. Bhaskaran, and I. Nesnas (2020). "Architecture Trades for Accessing Small Bodies with an Autonomous Small Spacecraft." *41st IEEE Aerospace Conference*.

Industrial Patents

- S. Papais, C. Cossette, L. Feng, and L. Ge (2026). *Object Detection Based On Filtering Of Environment Queries*. Patent filed 2025/09/19.

Selected Conference Presentations

- S. Papais. and S. Waslander. (2024). "Improving Temporal Reasoning for 3D Perception." *2024 Toronto Robotics Conference*.
- S. Papais., J. Hamel, and A. Nagaty (2021). "Crater-Based Absolute Navigation Demonstration." *2021 Canadian Lunar Workshop*.

TECHNICAL LEADERSHIP EXPERIENCE

Autonomous Driving Software Lead

07/2021 – 06/2023 aUToronto

- Led 30+ students to develop & deploy autonomous perception software.
- Integrated state-of-the-art ML multi-sensor detection and tracking models.
- Collected and curated dataset, optimized inference, and ran test campaigns.

📌 "aUToronto wins AutoDrive Challenge sequel." University of Toronto (2022).

Director, Technical Lead

09/2016 – 09/2018 McGill Rocket Team

- Managed a team of 100+ students to design, manufacture, and launch 6 sounding rockets up to 30,000 feet, and win an international competition.

📌 "McGill Rocket Team Soars to 1st at Spaceport America." McGill U. (2018).

AWARDS

- 🎓 **NASA JPL Research Fellowship 2019**
Awarded temporary research position.
- 🎓 **Elvie Smith Award 2019**
\$5k for contributions to aerospace.
- 🎓 **NSERC Research Exper. Award 2018**
\$5k to conduct R&D project.
- 🎓 **Next Generation Scholarship 2018**
\$3k for leadership and academics, by Ordre des Ingénieurs du Québec.
- 🎓 **McCaig Engineering & John Howard Ambrose Scholar 2016**
\$2k for 2015 academic performance.
- 🏆 **Forces Avenir Science Award 2018**
\$2k for community impact of Montreal Space Symposium.
- 🏆 **Director of the Year 2017**
McGill EUS recognition for new initiatives as funding director.

AFFILIATIONS

Vector AI Institute

Affiliate Researcher 2020 – Present

Computer Vision Foundation (CVF)

Member 2020 – Present

Inst. of Aero. & Astronautics (AIAA)

Senior Member 2018 – Present

Inst. of Electrical and Electronics Engineers (IEEE)

Member 2016 – Present

SKILLS

Machine Learning:

Representation Learning

Vision Transformers

3D Perception

Sensor Fusion

Object Detection

Object Tracking

Motion Forecasting

Self-Supervised

Foundation Models

Spatiotemporal Reasoning

VLMs

Robotics:

Control

Planning

Simulation

Safety-Critical Systems

Real-Time Systems

Technical:

PyTorch

JAX

TensorFlow

Python

C++

Docker

ROS

Linux

Distributed & Parallel Computing